NATIONAL AI HACKATHON 2026

Mind Reading Using Machine Learning and EEG

Team : KITKAT

Participants:
İrem Çiloğlu
Turgut Canberk Diner

Rank 102 Videos from 3 s of EEG

6	17	3 s	1,000	102
subjects	electrodes	per video	training videos	new test videos

1 · Video
predict the expected EEG

→

2 · Real EEG
wave + rhythm vectors

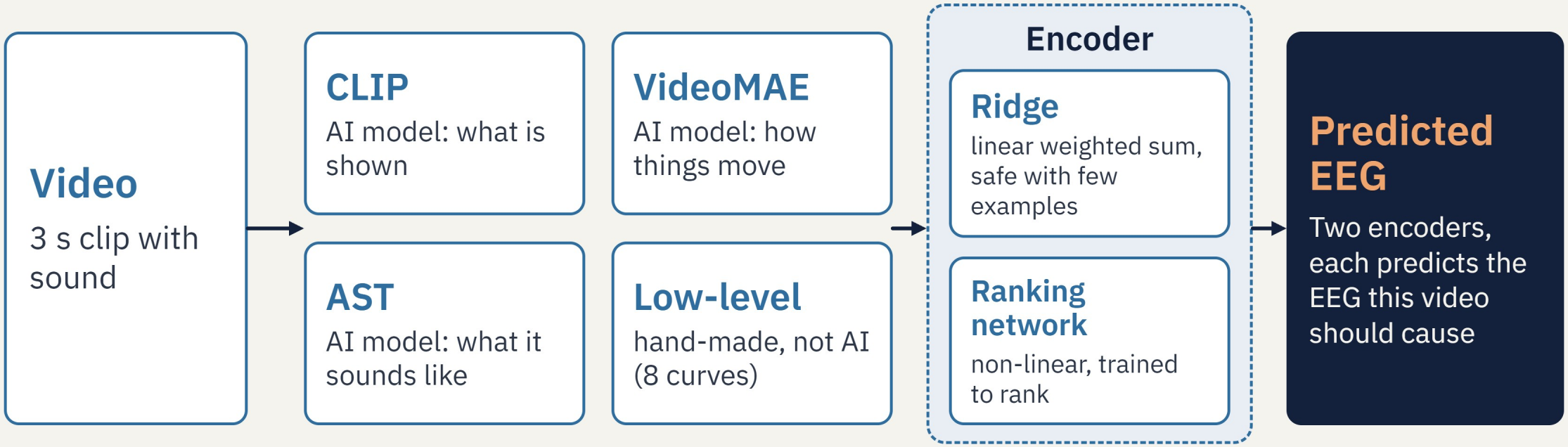
→

3 · Match
ranked list for subjects 5 & 6

How we measure: we never train or tune on test data. We tune only on training videos; subjects 1–4 are used only for reporting.

Our task: for each test EEG recording we rank 102 new videos. Subjects 5 and 6 are the ones we submit.

Step 1 · Video → Expected Brain Response



Why two encoders? Ridge is the stable linear model standard in neuroscience (Naselaris et al., 2011); the network adds non-linear effects and is trained for ranking. They make different mistakes, so together they score higher

Step 2 · Real EEG: Two Views

EEG
17 electrodes, 3 s

→

Clean
100 Hz · ZCA (Zero-phase Component Analysis) whitening: decorrelates the electrodes, keeps each channel close to its original

A · Wave

first 1 s: the brain's direct response

- **Why 1 s?** Longer windows were worse (3 s: 20.9% → 14.4% top-5). The dataset paper (Gifford et al.) finds video features predict EEG only up to ~1.5 s.
- **Why 100 Hz?** It holds frequencies up to fmax = fs / 2 = 50 Hz. We tried 25–500 Hz: no real difference.

Vector: 17 × 100 = 1,700 numbers

B · Rhythm Power

all 3 s: how strong each rhythm is

Delta 1–4 · **Theta** 4–8 · **Alpha** 8–13 · **Beta** 13–30 · **Low gamma** 30–45 Hz

- **How:** filter each band, square it (power), average per 0.5 s, take the log.
- **Why:** the exact wave fades after ~1 s, but rhythm strength still carries signal.

Vector: 17 × 5 × 6 = 510 numbers

Top-5 on held-out training videos, tuning splits, all 6 subjects. ZCA: Kessy et al., 2018.

Step 3 · Match and Rank

Step 1
video → predicted EEG

→

Step 2
real EEG: wave
+ rhythm

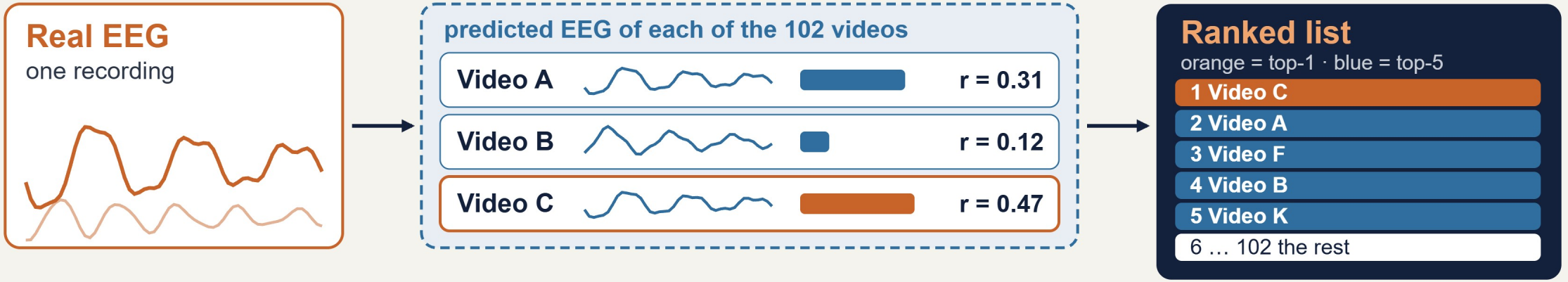
→

Match
correlate, hub fix

→

Result
102 videos ranked

- **Compare:** for each of the 102 videos, we compare its predicted EEG with the real EEG.
- **Score:** the more similar, the higher the score. Ridge and the network each give a score, and we add them up.
- **Rank:** videos that look similar to every recording get a lower score. The highest score is ranked first.



Results

19.9%
top-1

53.7%
top-5

4.6×
organisers' baseline

How We Know It Is Real

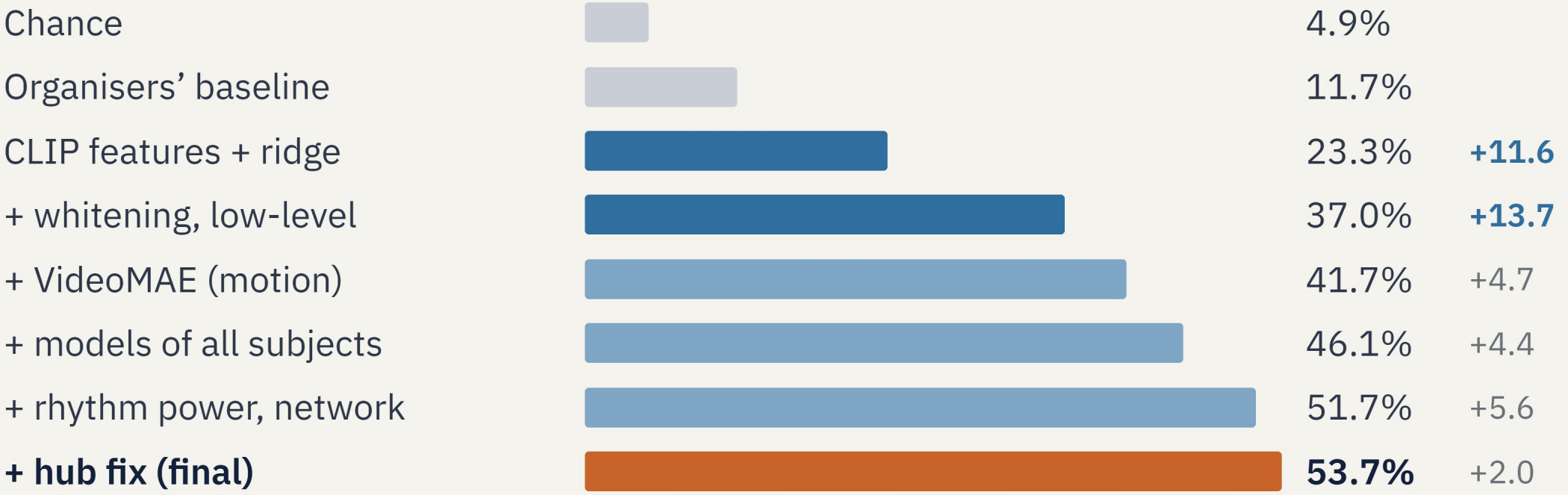
Tuned on training only
every choice made on held-out training videos

Fresh splits: 8.4% / 27.0%
never used for any choice; harder task

Shuffled labels: 1.0% / 4.9%
exactly chance, so nothing leaks

Top-1 / top-5. Test sets of subjects 1–4 (408 recordings), used for reporting only. Subjects 5 and 6: submitted rankings.

Where the 53.7% Came From



Biggest jumps: better input
AI video features and clean EEG gave 25 of the 42 points

Small steps are noise
the hub fix adds only +0.7 on fresh splits

Top-5 on the test sets of subjects 1–4; each bar adds one step. Each bar is one run, so steps of about 2 points are within noise.

What Did Not Help, and Why

Whitening Strength

no change



Whitening matters, its exact strength does not. We kept 0.3.

PCA Size

no change

64 | **128** | 256 | 512 — 128 was best

Ridge already shrinks weak features, so cutting more of them adds nothing.

Banded Ridge

worse



It learned weights close to ours (×0.7 to ×1.2), so ours were already right.

Network That Copies EEG

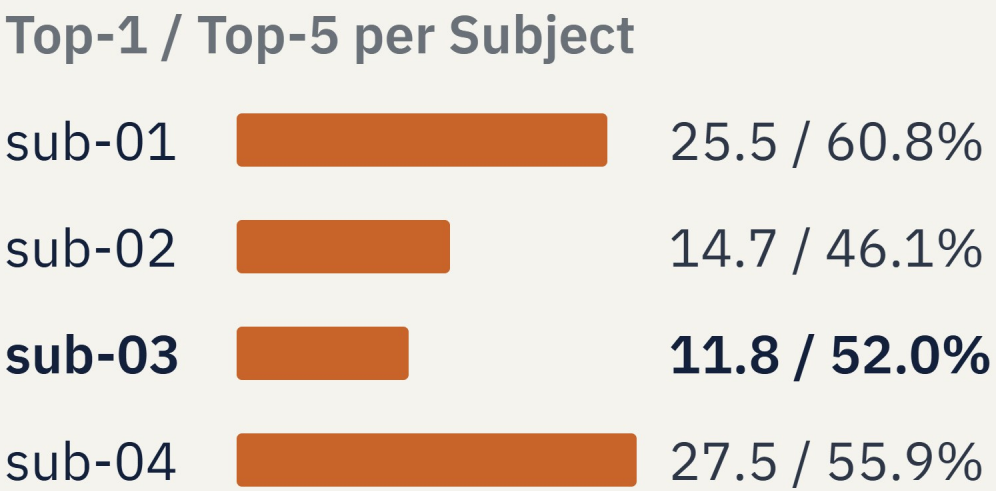
worse



The goal matters more than the network: training to rank is what helps.

Top-5 on held-out training videos, 3 tuning splits, all 6 subjects.

Where It Fails



Close, Not First

sub-03: top-5 as usual, rarely first

Hard Cases Stay Hard

fixes help near-misses only

Train ≠ Test

test EEG is cleaner (24 vs 6 views)

Brain and AI: What Our Results Say

Similar

AI Features Predict the Brain
4.6× the organisers' baseline

Middle Vision Layers Fit Best
better than the final layer

Motion Drives Both
motion features weigh most

Different

Better AI ≠ More Brain-Like
a stronger model didn't help

The Brain Has a Slow State
rhythms over seconds

Brains Vary, Models Don't
by viewing and by person

Lessons Learned

Most Tuning Is Noise

Keep a change only if it wins on every split.

New Evidence Beats Tuning

Rhythms and a ranking network.

Brain Science Guides Design

Fast response first, slow rhythms after.